

TAGISAN (`tgs`)

Sovereign Autonomous Agent Architecture

The Definitive Master Class Course & Production Blueprint

Dual-Brain Inference • BEAM/OTP Actors • Gleam • Google OAuth • AgentShield • 500 MCP Plugins • 150 Scenarios

Version:	3.0 (Dual-Brain / BEAM-OTP / Gleam / MCP-500)
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Engine Core:	tagisan-rs (Rust 1.78+ / Tokio Async Reactor / BEAM / Gleam)
Target Audience:	Principal Architects, SREs, DevSecOps, AI Engineers, Quantitative Traders
Repository:	https://github.com/CharleGutierrez/tagisan

Module 1: Dual-Brain Hybrid Inference & Google Cloud Integration

Tagisan combines local tensor inference with cloud frontier models into a unified, zero-cost-first failover engine.

- **Dual-Brain Architecture:** Local Ollama (Qwen2.5-Coder, SmolLM2) executes free daily tasks; Google Cloud CCPA gateway executes complex multi-step reasoning.
- **Google Web OAuth (AntiGravity CCPA Gateway):** Connects directly to Google's internal CloudCode Platform Assistant endpoint (daily-cloudcode-pa.googleapis.com) using OAuth 2.0 with a proactive 90-second token auto-refresh.
- **Model Fleet:** gemini-2.5-flash (default fast), gemini-3.1-pro-low (deep reasoning), gemini-2.5-flash-lite (sub-300ms latency), and gemini-3.6-flash-medium.
- **Graceful Logout Fallback:** Running tgs auth logout gemini resets the provider to auto, automatically falling back to local Ollama without errors.

```
tgs ask -m pro "Design a zero-downtime database migration"
tgs ask -m lite "Summarize active SRE incident alerts"
tgs auth status
```

Module 2: Polyglot Concurrency: Native Gleam & BEAM/OTP Actor Runtime

Tagisan integrates an Erlang/Elixir BEAM OTP actor runtime and a native Gleam compiler for crash-proof, distributed execution.

- **Actor Process Isolation:** Every task executes in an independent BEAM process with an isolated heap, preventing memory leaks and cross-thread corruption.
- **Supervisor Trees:** Implements one_for_one, one_for_all, and rest_for_one strategies with automatic restart telemetry.
- **Binary ETF Codec:** Full support for Erlang External Term Format (ETF 131) serialization at over 800,000 messages/second.
- **Type-Safe Gleam:** Statically typed Gleam microservices with algebraic data types and exhaustive pattern matching guarantee zero unhandled runtime panics.

Module 3: Command Ergonomics & Dialectical Debate (MoA)

Tagisan provides a unified CLI suite and Hegelian Dialectical Debate to eliminate hallucinations.

- **Hegelian Debate Consensus:** Spawns a Thesis Proposer, Antithesis Challenger, Synthesis Reconciler, and Judge to rigorously audit decisions.
- **Interactive TUI Chat:** High-speed terminal chat (tgs chat) with streaming reasoning tokens and token cost calculation.

```
tgs debate --proposer "Adopt Rust for core ledger" --challenger "Evaluate hiring velocity" --rounds 3
```

Module 4: AgentShield Cyber Defense & Landlock Sandboxing

Security is enforced at the kernel and Abstract Syntax Tree (AST) level prior to command execution.

- **Pre-Execution AST Interception:** Scans tool calls and shell arguments for destructive tokens (rm -rf /, fork bombs, disk formatters).
- **Linux Landlock LSM Sandboxing:** Unprivileged kernel jails restrict subprocess read/write access strictly to designated workspaces.
- **Credential Guard:** Automatically intercepts and blocks unauthorized access to ~/.ssh/id_rsa, .env, and cloud secrets.

Module 5: PILOT Just-In-Time Multi-Tier Memory

Multi-tier cognitive memory architecture with sub-millisecond relevance routing:

- **Tier 1 (Working Memory):** Tokio in-memory buffer holding active task context (< 1ms).
- **Tier 2 (Episodic Memory):** SQLite / DuckDB session history with transactional rollback (< 5ms).
- **Tier 3 (Semantic Memory):** High-dimensional vector database (Qdrant / SQLite-vec) with Reciprocal Rank Fusion (RRF) combining dense and BM25 sparse indices (< 15ms).

Module 6: The 500-Skill Sovereign Ecosystem (RFC-004)

All skills adhere to the RFC-004 package standard with YAML frontmatter specifications.

- Dynamic ingestion, automatic lifecycle validation, and sandboxed polyglot execution (Rust, Python, Gleam).
- Seamless chaining of domain skills in complex multi-stage autonomous missions.

Module 7: Model Context Protocol (MCP 500) Enterprise Hub

Tagisan serves as both an MCP client and an MCP server (tgs serve-mcp):

- **500 Verified Plugins:** Sourced from non-GitHub enterprise registries across 10 strategic domains (Databases, Cloud, DevOps, Security, APM, Productivity, SaaS/FinTech, AI, Vector DBs, Search).
- **Transports:** Full support for Stdio and SSE transports with JSON-RPC 2.0 communication.

Module 8: Domain Copilots & Cyber-Physical Digital Twins (VELLA)

VELLA brings real-time deterministic cyber-physical modeling to LLM agents:

- **SCADA / IoT:** Modbus TCP and OPC-UA sensor telemetry sync, automated alarm actions, and valve control.
- **Quantitative Finance:** High-frequency forex tick spreads, Value-at-Risk (VaR) Monte Carlo simulations, and margin de-leveraging.
- **Aerospace:** SGP4 Two-Line Element (TLE) satellite orbit propagation and ground station visibility pass forecasting.
- **Bioinformatics:** FASTA/FASTQ sequence alignment (Smith-Waterman) and VCF mutation filtering.

Module 9: Enterprise Hardening & Air-Gapped Operations

Designed for sovereign government, defense, and enterprise deployments:

- **Air-Gapped Isolation:** Setting TAGISAN_LOCAL_ONLY=1 guarantees 100% offline execution using local Ollama tensors.
- **OpenTelemetry Observability:** Structured JSON logging of every token, turn latency (ms), AgentShield verdict, and exact USD cost.

Module 10: The Definitive 150 Production Scenarios

This catalog details 150 concrete, production-proven scenarios across 10 major enterprise engineering domains. Every scenario demonstrates how Tagisan's dual-brain, BEAM actor runtime, AgentShield security, and MCP ecosystem solve mission-critical challenges in modern environments.

Domain 1–15: ■■ Cloud Architecture, SRE & Kubernetes Operations

Scenario 1: Kubernetes Pod CrashLoopBackOff Auto-Diagnosis and Rollback
Capabilities: AgentShield, Kubernetes MCP, Local Ollama Fallback
tgs run --skill "k8s-pod-diagnostics" "Analyze CrashLoopBackOff in pod auth-svc-78bd in namespace prod"
1. Queries kube-apiserver for pod events and previous container logs. 2. Identifies OOMKilled condition caused by memory leak in v2.4.1. 3. Checks Helm release history and issues safe rollback command with AgentShield validation.
Outcome: Pod rolled back to v2.4.0 within 45s; zero manual SRE downtime.
Scenario 2: Multi-Region Terraform Drift Detection & Plan Reconciliation
Capabilities: Terraform MCP, Swarm MoA (Proposer + Auditor)
tgs run "Compare active AWS us-east-1 and eu-central-1 infrastructure against main.tf state and heal drift"
1. Executes `terraform plan -detailed-exitcode` across both regions. 2. Discovers manually modified security group allowing inbound 0.0.0.0/0 on port 22. 3. Generates reconciliation PR and auto-applies least-privilege CIDR rules.
Outcome: Security group restored to VPC-only CIDR without service disruption.
Scenario 3: Istio Service Mesh Mutual TLS Certificate Expiry Auto-Rotation
Capabilities: AgentShield, Bash Sandboxing, OpenSSL Parser
tgs run "Audit all Istio mTLS workload certificates expiring within 7 days and trigger Citadel rotation"
1. Scans Envoy secret dumps on 120 mesh sidecars. 2. Flags 4 certificates with under 48 hours remaining due to failed SDS sync. 3. Triggers envoy SDS reload and verifies handshake success using TLS probe.
Outcome: 100% mTLS certificate renewal completed with zero dropped connections.
Scenario 4: Prometheus Alert Fatigue Suppression & Root Cause Clustering
Capabilities: PILOT Vector Memory, Prometheus MCP, Gemini 3 Flash
tgs run "Cluster 450 firing Prometheus alerts from incident #8821 to identify primary root cause"
1. Ingests raw Alertmanager JSON payload over stdio MCP. 2. Uses PILOT semantic similarity to group 442 cascading downstream HTTP 504 alerts. 3. Isolates primary failure: Redis connection pool starvation on primary leader node.
Outcome: Root cause pinned in 1.4s; on-call engineer alerted to 1 actionable ticket instead of 450.
Scenario 5: Zero-Downtime PostgreSQL Schema Migration with PgBouncer Pooling
Capabilities: PostgreSQL MCP, Hegelian Dialectical Debate
tgs debate --proposer "Add NOT NULL column user_uuid to 50M row users table" --challenger "Prevent table locks"

1. Proposer suggests ALTER TABLE ADD COLUMN.
2. Challenger proves ALTER TABLE takes ACCESS EXCLUSIVE lock stalling all web requests.
3. Synthesis crafts 3-step zero-lock migration: ADD COLUMN NULLABLE -> BACKFILL BATCHES -> ADD VALIDATED CONSTRAINT.

Outcome: 50M row migration executed with 0ms query lock latency.

Scenario 6: AWS IAM Least-Privilege Policy Pruning & Overprivileged Role Remediation

Capabilities: AWS IAM MCP, AgentShield AST Interceptor

tgs run "Analyze CloudTrail 90-day activity for role app-backend and remove unused wildcard permissions"

1. Parses CloudTrail access events matching AssumeRole for `app-backend`.
2. Detects `s3:\*` and `dynamodb:\*` wildcards with zero DeleteBucket or DropTable events.
3. Generates scoped JSON IAM policy granting read/write on exact bucket ARNs.

Outcome: Attacking surface reduced by 88% while preserving all production workloads.

Scenario 7: Chaos Engineering Injector: Automated Network Latency and Pod Eviction

Capabilities: Chaos Mesh MCP, BEAM Actor Supervisor Tree

tgs run "Inject 200ms latency on payment-gateway namespace for 10 minutes and audit circuit breakers"

1. Spawns BEAM supervisor actor to monitor application error budget.
2. Applies Chaos Mesh network latency CRD to egress routes.
3. Verifies resilience: Resilience4j circuit breaker opens and falls back to cached payments.

Outcome: Payment failure rate stayed under 0.01%; recovery validated automatically.

Scenario 8: Cloud Cost Anomaly Hunter: Idle EBS Volumes & Zombie EKS Clusters

Capabilities: Cloud Cost MCP, SQLite Episodic Memory

tgs run "Scan AWS account for unattached gp3 volumes and idle dev EKS clusters active > 30 days"

1. Queries AWS EC2/EKS APIs for volume state and worker node CPU utilization.
2. Discovers 14 unattached EBS volumes (8.4 TB) and 2 idle test clusters consuming \$2,800/mo.
3. Takes snapshots of unattached volumes, archives metadata, and issues termination requests.

Outcome: Immediate \$33,600 annual cloud savings realized safely.

Scenario 9: Distributed Tracing Span Bottleneck Pinpointer with OpenTelemetry

Capabilities: Jaeger/OpenTelemetry MCP, Gemini 3.1 Pro Low

tgs run "Analyze 1,000 p99 traces for /checkout endpoint and isolate latency spike sources"

1. Fetches high-latency trace spans from Jaeger collector.
2. Traverses DAG call graph across 12 microservices.
3. Detects unindexed SQL query inside coupon validation service executing 48 repeated queries per request.

Outcome: N+1 query discovered; patch generated reducing checkout latency from 3.2s to 120ms.

Scenario 10: Automated Ingress NGINX CVE Mitigation and Lua Security Rule Injection

Capabilities: AgentShield, Kubernetes Secret Engine, Fast-Patching

tgs run "Scan NGINX ingress controller against CVE-2023-5043 and apply ingress annotation mitigations"

1. Evaluates ingress controller image tag against NVD vulnerability database.
2. Detects vulnerability in custom snippet execution.
3. Patches ingress controller ConfigMap to disable custom snippets and injects WAF regex filter.

Outcome: Zero-day ingress exploit blocked across 24 public domains in 3 minutes.

Scenario 11: Multi-Cloud Failover Orchestration (AWS us-east-1 to GCP us-central1)

Capabilities: Route53 MCP, Cloud DNS MCP, Swarm Consensus

tgs run "Simulate AWS us-east-1 regional blackhole and execute DNS failover to GCP backup cluster"

1. Probes synthetic healthcheck endpoint in AWS us-east-1; confirms 100% packet loss.
2. Updates Route53 latency-based routing records to point traffic to Google Cloud GKE ingress IP.
3. Verifies database read-replica promotion on GCP Cloud SQL.

Outcome: Full application traffic rerouted to GCP with total RTO under 90 seconds.

Scenario 12: Kubernetes Horizontal Pod Autoscaler (HPA) Predictive Scaling with Ollama

Capabilities: Local Ollama (Qwen2.5-Coder), Prometheus Metrics API

tgs run "Analyze 30-day traffic cyclicity and generate predictive HPA cron schedules for Black Friday"

1. Extracts hourly request-per-second timeseries from Prometheus.
2. Executes local Ollama autoregressive analysis to predict upcoming peak traffic bursts.
3. Deploys KEDA (Kubernetes Event-driven Autoscaling) CronScaledObject to scale pods 15m prior to load.

Outcome: Zero 503 throttling during flash sale spikes.

Scenario 13: Cloudflare Edge Worker Deployment & Cache Purge Pipeline

Capabilities: Cloudflare MCP, Bun Fast-Runtime Tooling

tgs run "Deploy Geo-IP routing Cloudflare Worker and purge edge cache for static asset bundles"

1. Validates TypeScript worker syntax using Bun runtime.
2. Publishes worker to Cloudflare edge network across 300+ PoPs.
3. Executes targeted cache purge for `/static/bundle.v2.js` via Cloudflare API token.

Outcome: Worker deployed globally in 2.1s with verified edge cache invalidation.

Scenario 14: GitOps ArgoCD Application Sync Failure Triangulation and Commit Healing

Capabilities: GitOps MCP, Dialectical Debate, Git Integration

tgs run "Investigate OutOfSync status on ArgoCD app payment-service and resolve manifest schema error"

1. Queries ArgoCD REST API for application diff; detects unrecognized field `autoscaling/v2beta1`.
2. Upgrades Kubernetes API version in deployment repository to `autoscaling/v2`.
3. Commits fix with verified GPG signature and triggers ArgoCD automated sync.

Outcome: ArgoCD status restored to Synced/Healthy in under 60 seconds.

Scenario 15: Automated Disaster Recovery Backup Verification and RTO/RPO Benchmarking

Capabilities: AWS S3 MCP, PostgreSQL Dump Engine, AgentShield

tgs run "Restore latest nightly DB backup to staging scratch cluster and measure exact RTO and data integrity"

1. Downloads encrypted pg_dump archive from S3 bucket with Landlock sandboxing.
2. Provisions temporary ephemeral PostgreSQL container and restores 180GB database.
3. Executes checksum row-count verification across critical financial ledger tables.

Outcome: RTO clocked at 18 minutes (SLA: 1 hour); RPO verified at 42 seconds; report signed.

Domain 16–30: ■■ Cybersecurity, DevSecOps & Incident Response

Scenario 16: Real-Time SOC Alert Triage & Phishing Email Header Forensics

Capabilities: Email Forensics MCP, AgentShield AST Scanner

tgs run "Analyze suspicious email header attachment ticket #9021 for domain spoofing and malicious payload"

1. Parses RFC 822 email headers; validates DKIM, SPF, and DMARC alignment.
2. Discovers failed SPF check from lookalike domain `paypa1.com`.
3. Extracts macro-enabled Excel attachment in memory and neutralizes reverse shell callout.

Outcome: Malicious sender IP blocked on enterprise Palo Alto firewall within 12 seconds.

Scenario 17: Automated Secret Exfiltration Prevention & Git History Scrubbing (BFG) Capabilities: Git History Engine, AgentShield Credential Guard tgs run "Detect leaked AWS_SECRET_ACCESS_KEY in git commit history and rewrite repository tree" 1. Performs high-speed regex and entropy scan over all 4,200 git commits. 2. Discovers exposed AWS secret key committed 3 months prior in deleted config file. 3. Invokes BFG repo-cleaner to purge blob, triggers force-push, and rotates AWS IAM access key. Outcome: Secret revoked on AWS IAM and completely scrubbed from git history.
Scenario 18: Linux Landlock LSM Kernel Sandboxing for Unverified Agent Tools Capabilities: Landlock LSM Kernel Interceptor, AgentShield tgs run --sandbox strict "Execute third-party data extraction binary and restrict file access to /tmp/scratch" 1. Configures Linux Landlock ruleset: blocks read/write to `/etc`, `/home`, `/root`. 2. Strips network capabilities (`CAP_NET_RAW`, `CAP_NET_ADMIN`). 3. Executes untrusted binary; intercepts attempt to read `/etc/passwd` with immediate SIGKILL. Outcome: Zero-trust sandbox contained exploit cleanly without kernel compromise.
Scenario 19: Mitigating SQL Injection Vulnerabilities in Legacy Codebases Capabilities: Static Analysis Engine, Dialectical Code Synthesis tgs run "Scan src/legacy_auth.php for SQL injection vectors and rewrite queries using PDO prepared statements" 1. Identifies string concatenation in `SELECT * FROM users WHERE user = '\$username'`. 2. Rewrites logic to use parameter binding with PDO. 3. Generates automated PHPUnit integration test verifying that ` ` OR '1'=1` fails authentication. Outcome: High-severity vulnerability remediated with automated regression tests.
Scenario 20: Reverse Engineering Obfuscated Malicious Bash Payloads with AgentShield Capabilities: AgentShield Deobfuscator, Local Ollama DeepSeek tgs run "Deobfuscate base64-encoded pipe-to-bash script intercepted on honeypot server" 1. Extracts nested base64, gzip, and rot13 layers in memory without shell execution. 2. Discovers persistence mechanism creating systemd cron service downloading cryptominer. 3. Outputs full IOC report including C2 IP addresses and file hashes. Outcome: Complete threat intelligence report generated and pushed to SIEM.
Scenario 21: Dynamic API Fuzzing and OpenAPI Specification Flaw Detection Capabilities: API Fuzzing MCP, Gemini 3 Flash tgs run "Perform property-based fuzz testing on /api/v1/orders endpoint using openapi.yaml spec" 1. Generates 50,000 edge-case payloads (boundary integers, null bytes, unicode emojis, oversized strings). 2. Uncovered unhandled 500 internal server error when sending negative quantity integer. 3. Submits pull request adding input validation constraint in Rust Axum controller. Outcome: Denial-of-Service vector eliminated before production deployment.
Scenario 22: Zero-Day Patch Synthesis for OpenSSL Buffer Overflows Capabilities: C/C++ Ast Engine, Swarm MoA (Security Auditor + C Expert) tgs debate --proposer "Synthesize safe boundary check patch for CVE-2022-3602 in libssl" --challenger "Verify ABI compatibility" 1. Audits punycode decoding routine in OpenSSL X.509 name parsing. 2. Identifies 4-byte stack overflow vulnerability on 32-bit platforms. 3. Crafts ABI-compliant patch with bounded length verification.

Outcome: Patch verified against OpenSSL regression test suite.
Scenario 23: MITRE ATT&CK; Mapping of Active Directory Lateral Movement Telemetry
Capabilities: Windows Event Log Parser, Vector Memory RRF
tgs run "Map EventID 4624 (Type 3) and 7045 spikes across domain controllers to MITRE ATT&CK; tactics"
1. Ingests 500,000 Windows Security Event logs from domain controllers. 2. Flags Pass-the-Hash pattern followed by remote PsExec service installation. 3. Correlates indicators to MITRE T1021.002 (SMB/Windows Admin Shares) and T1569.002 (Service Execution).
Outcome: Compromised workstation isolated from Active Directory domain in 4 minutes.
Scenario 24: Automated Container Image Vulnerability Triaging (Trivy + SBOM Matching)
Capabilities: Trivy MCP, Syft SBOM Generator, Gemini 2.5 Flash
tgs run "Scan container registry image api-gateway:v3.2 for CRITICAL CVEs and filter non-exploitable packages"
1. Generates CycloneDX Software Bill of Materials (SBOM) using Syft. 2. Matches vulnerabilities against live runtime call-graph. 3. Filters out 18 CVEs in unused test binaries; flags 1 actionable CVE in active libxml2 parser.
Outcome: Base image upgraded to Alpine 3.20; vulnerability count dropped from 19 to 0.
Scenario 25: Ransomware Behavior Detection in Distributed NFS/Ceph Storage Nodes
Capabilities: AgentShield I/O Watcher, SCADA/IoT Twin Engine
tgs run "Monitor storage node /mnt/data for mass file extension renaming and entropy spikes"
1. Samples file modification rates; detects 1,400 files/sec being renamed to `.locked`. 2. Shannon entropy analysis confirms encrypted high-entropy payload substitution. 3. Immediately revokes compromised NFS client IP and freezes Ceph volume snapshot.
Outcome: Ransomware spread halted in 1.8s; 99.4% of corporate data preserved via immediate snapshot.
Scenario 26: Web Application Firewall (WAF) Dynamic Rule Generation from Access Logs
Capabilities: Cloudflare WAF MCP, Regular Expression Synthesizer
tgs run "Analyze 403/500 spikes from access.log and deploy Cloudflare WAF custom rule blocking scraper botnet"
1. Identifies distributed botnet rotating through 400 residential proxies with common TLS fingerprint. 2. Discovers unique user-agent header casing irregularity: `Mozilla/5.0 (Windows NT 10.0; WOW64; x64)`. 3. Deploys Cloudflare WAF rule combining JA3 fingerprint and header pattern.
Outcome: Bot traffic dropped from 94% to 0.01% without impacting legitimate users.
Scenario 27: Memory Corruptor & Race Condition Hunter via Rust ThreadSanitizer
Capabilities: Rust Cargo Engine, Valgrind / TSan Profiler
tgs run "Run cargo test with -Zsanitizer=thread on high-throughput actor mailbox and fix data race"
1. Executes multithreaded stress test under TSan instrumentation. 2. Flags unsynchronized read/write on atomic reference counter in custom lock-free ring buffer. 3. Replaces relaxed memory ordering (`Ordering::Relaxed`) with acquire-release semantics (`Ordering::AcqRel`).
Outcome: Race condition eliminated with zero benchmark throughput penalty.
Scenario 28: Cloud Security Posture Management (CSPM) CIS Benchmark Automated Remediation
Capabilities: AWS Security Hub MCP, AgentShield
tgs run "Audit AWS account against CIS Benchmark v1.4 and auto-remediate unencrypted S3 buckets"
1. Evaluates all 82 S3 buckets across 4 AWS regions. 2. Flags 3 legacy buckets missing Default Encryption and Public Access Block. 3. Applies AES-256 (SSE-S3) encryption and enables bucket policy enforcing HTTPS only.

Outcome: CIS Benchmark compliance score elevated from 78% to 98%.
Scenario 29: Privilege Escalation Path Mapping in Kubernetes RBAC Graph
Capabilities: Petgraph Engine, Kubernetes RBAC MCP
tgs run "Build directed graph of all ServiceAccounts, Roles, and Bindings to discover escalation to cluster-admin"
1. Ingests all ClusterRoles, Roles, and RoleBindings into in-memory Petgraph. 2. Executes Dijkstra shortest-path search from default namespace service accounts to `cluster-admin`. 3. Discovers service account with `create` permission on `pods/exec` allowing privilege escalation.
Outcome: Overprivileged RoleBinding removed; escalation vulnerability closed.
Scenario 30: Post-Mortem Incident Timeline Generation and Executive Debrief Synthesis
Capabilities: PILOT Episodic Memory, Gemini 3.1 Pro Low
tgs run "Synthesize complete post-mortem timeline from Slack incident channel and PagerDuty alert logs"
1. Aggregates timestamps from PagerDuty, Slack war-room channel, and GitHub deployment commits. 2. Structures timeline down to minute precision: Detection (02:14), Triage (02:18), Mitigation (02:41). 3. Synthesizes executive summary, Root Cause Analysis (RCA), and 5 Preventative Action Items.
Outcome: Executive-ready Post-Mortem document published to Confluence in markdown.

Domain 31–45: ■ Full-Stack & Systems Software Engineering

Scenario 31: Legacy Monolith to Microservices Domain-Driven Refactoring
Capabilities: AST Refactoring Engine, Swarm MoA Architecture Team
tgs debate --proposer "Extract billing domain from monolithic Django app into Axum Rust service" --challenger "Maintain transactional consistency"
1. Proposer maps Django ORM models (`Invoice`, `Payment`, `Subscription`). 2. Challenger highlights distributed transaction risks and dual-write anomalies. 3. Synthesis crafts Outbox Pattern architecture using Kafka CDC events.
Outcome: Clean microservice boundary created with zero lost billing transactions.
Scenario 32: Autonomous Pull Request Review: Code Quality, Complexity & Test Coverage
Capabilities: GitHub MCP, AST Parser, Gemini 2.5 Flash
tgs run "Review pull request #142 in repo frontend-core for cyclomatic complexity and missing unit tests"
1. Ingests git unified diff across 22 changed files. 2. Identifies cyclomatic complexity of 34 in nested authentication reducer. 3. Writes constructive inline GitHub review comments and generates Jest test covering edge cases.
Outcome: Review posted to GitHub in 14s; PR author merged proposed refactoring.
Scenario 33: High-Throughput Async Tokio Reactor Optimization in Rust Web Services
Capabilities: Rust Compiler Engine, Tokio Console Profiler
tgs run "Profile Tokio task scheduling in src/network/reactor.rs and eliminate async blocking calls"
1. Inspects async tasks using Tokio tracing instrumentation. 2. Discovers synchronous `std::fs::read` executing inside high-frequency worker loop, stalling thread pool. 3. Refactors to `tokio::fs::read` and offloads heavy crypto hashing to `tokio::task::spawn_blocking`.
Outcome: Request throughput increased by 420% with p99 latency dropping from 80ms to 4ms.
Scenario 34: React to Next.js 15 Server Components Migration with Zero Regression

Capabilities: Bun Toolchain, TypeScript AST Rewriter tgs run "Migrate client-side React SPA in src/pages/dashboard to Next.js 15 App Router Server Components" 1. Analyzes React component tree to separate interactive state (`use client`) from pure render trees. 2. Converts client-side `useEffect` data-fetching to async Server Components with streaming Suspense. 3. Verifies zero bundle size regression using Next.js bundle analyzer. Outcome: First Contentful Paint (FCP) improved from 2.4s to 0.3s; JS client bundle reduced by 62%.
Scenario 35: Database Query N+1 Identification and ORM Eager-Loading Synthesis Capabilities: SQL Parser, Database MCP, Gemini 3 Flash tgs run "Profile Hibernate ORM queries on GET /api/v1/organizations and eliminate N+1 select queries" 1. Ingests query execution logs; detects 1 initial query followed by 850 individual child queries. 2. Rewrites JPA query using `JOIN FETCH o.members m JOIN FETCH m.permissions`. 3. Verifies database query count reduced from 851 to 1 single index-backed query. Outcome: Endpoint execution time dropped from 4,200ms to 28ms.
Scenario 36: Cross-Platform GUI Tooling with Ratatui & Crossterm TUI Capabilities: Rust Compiler Engine, Crossterm Simulator tgs run "Implement interactive Terminal UI in Rust using Ratatui to monitor real-time cluster node health" 1. Synthesizes full Ratatui application state, layout splits, and event-handling loop. 2. Renders ASCII sparklines, gauge bars for CPU/RAM, and color-coded table of running pods. 3. Implements non-blocking keyboard navigation and terminal resize listeners. Outcome: Zero-dependency terminal monitor compiled into a single 4.2MB binary.
Scenario 37: gRPC Protobuf Contract Backward-Compatibility Verification Capabilities: Protobuf Engine, Buf CLI MCP tgs run "Compare updated proto/billing.proto against production v1.2 schema for breaking changes" 1. Compiles proto definitions using `buf breaking --against git://...`. 2. Flags deletion of field #4 (`string billing_zip`) as breaking wire-format change for mobile clients. 3. Recommends marking field as `reserved 4;` and adding new field #5. Outcome: Breaking wire protocol change caught and prevented prior to release.
Scenario 38: WebAssembly (Wasm) Micro-Module Compilation and Sandbox Embedding Capabilities: Wasmtime Runtime Engine, Rust Wasm Toolchain tgs run "Compile image transformation algorithm in src/filters/ into Wasm and embed via Wasmtime" 1. Compiles Rust source to `wasm32-wasi` target with optimization flags. 2. Provisions Wasmtime engine with strict fuel-metering and memory limit of 64MB. 3. Executes transformation in sandbox; benchmarks execution against native speeds. Outcome: Isolated plugin execution achieved at 94% of native performance.
Scenario 39: Native C/C++ Memory Leak Profiling with Valgrind and ASan Capabilities: Clang/LLVM Engine, AddressSanitizer (ASan) tgs run "Compile C++ packet parser with -fsanitize=address and isolate heap-use-after-free" 1. Executes packet ingestion test harness under ASan instrumentation. 2. Catches heap-use-after-free on socket buffer deallocation in worker thread. 3. Rewrites buffer ownership using `std::unique_ptr` and verified leak-free report. Outcome: Critical memory vulnerability fixed with zero Valgrind errors.
Scenario 40: Continuous Benchmarking and Performance Regression Gatekeeper

Capabilities: Criterion.rs Engine, GitHub Actions MCP tgs run "Run criterion benchmark suite comparing current commit against main branch" - Executes 10,000 iterations of JSON serialization benchmark. - Statistical analysis detects a +14.2% regression in parsing floating-point numbers. - Identifies replacement of `fast-float` crate with slower standard library parser; reverts change. Outcome: Performance regression blocked from entering release branch.
Scenario 41: Automated Documentation Generation with OpenAPI and Typed Interfaces Capabilities: OpenAPI Spec Engine, Gemini 2.5 Flash tgs run "Extract OpenAPI 3.1 specification directly from Axum router handlers in src/api/" - Traverses Rust AST extracting route paths, input request structs, and HTTP response codes. - Generates comprehensive `openapi.json` with accurate JSON schemas and docstrings. - Verifies Swagger UI rendering and mock server response matching. Outcome: Production API documentation automatically kept in 100% sync with source code.
Scenario 42: Frontend Internationalization (i18n) Extraction and Automated Translation Capabilities: i18n Extraction Engine, Multi-Language LLM tgs run "Extract all hardcoded English strings from React components into locales/en.json and translate to ja/es/de" - Scans JSX components for raw string literals outside of translation hooks. - Generates keyed i18n JSON dictionary and replaces code with `t('key')` calls. - Produces high-fidelity translations in Japanese, Spanish, and German with context preservation. Outcome: Enterprise i18n rollout executed across 84 screens in 3 minutes.
Scenario 43: WebSocket Heartbeat and Distributed Connection Pool Resiliency Capabilities: Tokio WebSocket Engine, Redis PubSub MCP tgs run "Design resilient WebSocket gateway handling 50,000 concurrent client connections with ping/pong keepalive" - Implements Tokio-tungstenite connection worker with heartbeat timeout of 30s. - Connects connection state to Redis cluster via PubSub broadcast. - Simulates network disconnection; verifies automated client reconnection without duplicate sessions. Outcome: Stable 50k connection pool maintained with sub-millisecond broadcast latency.
Scenario 44: Legacy Python 2 to 3.12 Polyglot Migration with Type Annotations Capabilities: Python AST Engine, Ruff Linter, uv Package Manager tgs run "Migrate legacy Python 2.7 data script to Python 3.12 with full typing and mypy validation" - Converts `print` statements, `xrange` to `range`, and unicode string encodings. - Adds PEP 484 type hints across all function signatures. - Runs `ruff` formatting and validates zero mypy type errors. Outcome: Legacy script modernized with 3.8x runtime speedup on Python 3.12.
Scenario 45: Build System Modernization (Make -> Cargo / Bun / Bazel) Capabilities: Build System Engine, Cargo / Bun / Bazel MCP tgs run "Convert complex 1,200-line Makefile into hermetic Bazel build targets with remote caching" - Analyzes dependency graph across C++, Rust, and TypeScript components. - Generates Bazel `WORKSPACE` and modular `BUILD.bazel` rules. - Validates reproducible build output and remote cache hit rate. Outcome: Clean build times reduced from 42 minutes to 3.5 minutes.

Domain 46–60: ■ Polyglot Compilation, Gleam & BEAM/OTP Actor Concurrency

Scenario 46: Compiling Type-Safe Gleam Micro-Services into BEAM Bytecode

Capabilities: Native Gleam Compiler Engine, BEAM VM

```
tgs run "Compile Gleam web service in src/gleam_app into BEAM bytecode and verify actor types"
```

1. Invokes native Gleam compiler; performs algebraic data type checking.
2. Confirms exhaustive pattern matching on all domain events.
3. Emits validated `.beam`` bytecode ready for distributed Erlang nodes.

Outcome: Zero compiler warnings; type-safe bytecode produced in 420ms.

Scenario 47: Erlang/Elixir BEAM Supervisor Tree Crash Isolation (one_for_one)

Capabilities: OTP Supervisor Engine, Fault Tolerance Simulator

```
tgs run "Simulate fatal divide-by-zero panic in worker actor #4 and verify OTP supervisor auto-restart"
```

1. Injects intentional panic inside running GenServer process.
2. BEAM supervisor intercepts crash; records crash report with stack trace.
3. Restarts failed worker with fresh state within 2 milliseconds without disturbing sibling workers.

Outcome: 'Let it crash' resilience verified; 99.999% uptime maintained.

Scenario 48: Binary ETF (External Term Format) Serialization for Cross-Process Interop

Capabilities: ETF Codec Engine, Rust-BEAM Bridge

```
tgs run "Serialize 100,000 nested telemetry records into Erlang External Term Format (ETF) in Rust"
```

1. Maps Rust struct hierarchy to Erlang atoms, tuples, lists, and binaries.
2. Encodes data using fast binary ETF codec format 131.
3. Sends payload to Elixir GenServer over Unix domain socket; verifies zero-copy decoding.

Outcome: ETF serialization clocked at 820,000 records/sec; 40% smaller than JSON.

Scenario 49: Building Resilient Fault-Tolerant Actor Mailbox Queues in Gleam

Capabilities: Gleam OTP Engine, Actor Mailbox Watcher

```
tgs run "Implement bounded actor mailbox queue in Gleam with backpressure and dead-letter queue"
```

1. Defines message type with timeout and priority tags.
2. Implements actor receiver loop dropping low-priority telemetry when mailbox exceeds 10,000 messages.
3. Routes dropped messages to persistent SQLite dead-letter queue for forensic replay.

Outcome: Actor process prevented from OOM crash under 50x network traffic surge.

Scenario 50: Hot Code Reloading on Live Elixir Nodes without Process Termination

Capabilities: BEAM Hot-Code Reloader, Elixir SDK

```
tgs run "Deploy updated payment_calc.ex module to live production BEAM cluster without dropping connections"
```

1. Compiles modified Elixir source to `.beam`` object.
2. Transmits module update to live Erlang runtime using `:code.load_binary/3``.
3. Existing processes smoothly transition to new code on next message loop iteration.

Outcome: Zero dropped socket connections; live production code hot-swapped in 15ms.

Scenario 51: Distributed GenServer Process Registry Clustering with Phoenix PubSub

Capabilities: Erlang Distributed Node Engine, Phoenix PubSub

```
tgs run "Cluster 3 BEAM nodes across VPC and verify global process lookup by customer UUID"
```

1. Establishes distributed Erlang clustering using EPMD and shared cookie.
2. Registers GenServer processes using `:global` and distributed Horde registry.
3. Dispatches message from Node A to customer process running on Node C transparently.

Outcome: Cluster unified with transparent multi-node message passing.

Scenario 52: Gleam Type System Algebraic Data Type (ADT) Pattern Matching Engine

Capabilities: Gleam Type System, Dialectical Code Synthesis

tgs run "Design comprehensive payment state machine in Gleam with compile-time unhandled case enforcement"

1. Defines `PaymentState` ADT: `Pending`, `Authorized`, `Captured`, `Refunded`, `Failed`.
2. Writes state transition function; compiler flags missing match on `Refunded` from `Pending`.
3. Resolves state transitions with strict mathematical proofs.

Outcome: Invalid payment state transitions rendered impossible at compile time.

Scenario 53: Cross-Language FFI Binding Generation (Rust napi-rs to Bun/Node)

Capabilities: Rust FFI Engine, Bun Fast-Runtime

tgs run "Generate high-performance Node-API (NAPI) bindings for Rust blake3 hashing engine in Bun"

1. Authors Rust napi-rs bridge wrapping parallel Blake3 multithreaded hashing.
2. Compiles `.node` native binary and TypeScript `.d.ts` definitions.
3. Benchmarks execution in Bun against native JS crypto: achieves 28x throughput improvement.

Outcome: Zero-overhead native binding integrated into TypeScript services.

Scenario 54: Erlang Mnesia Distributed In-Memory Database Transaction Coordination

Capabilities: Mnesia Database Engine, OTP Actor System

tgs run "Configure Mnesia replicated ram_copies table across 3 nodes with ACID transaction guarantees"

1. Initializes Mnesia schema on 3 distributed nodes.
2. Creates distributed table with `ram\_copies` and dirty read caching.
3. Executes 5,000 atomic transactions per second with automated partition split-brain recovery.

Outcome: High-speed in-memory state replication verified with zero data corruption.

Scenario 55: OTP rest_for_one Supervisor Strategy for Dependent Pipeline Subsystems

Capabilities: OTP Supervisor Engine, System Health Monitor

tgs run "Configure rest_for_one supervisor managing DatabaseConn -> CacheSync -> WebRouter"

1. Establishes startup dependency order: DB, Cache, Router.
2. Simulates crash in CacheSync component.
3. Supervisor restarts CacheSync and downstream WebRouter while keeping DatabaseConn alive.

Outcome: Targeted subsystem recovery achieved without restarting database connection pools.

Scenario 56: Gleam Web Framework (Wisp) Production API Deployment

Capabilities: Gleam Compiler, Wisp / Mist HTTP Engine

tgs run "Build and benchmark a Gleam Wisp REST API handling JSON requests with Mist HTTP server"

1. Authors Gleam route handlers with middleware for request logging and CORS.
2. Decodes JSON requests using typed Gleam decoders.
3. Benchmarks Mist HTTP server: clocks 110,000 requests/sec with 0.8ms average latency.

Outcome: Lightweight, crash-proof REST microservice deployed successfully.

Scenario 57: Actor Deadlock and Message Flood Detection under High Network Load

Capabilities: BEAM Observer Engine, AgentShield

tgs run "Monitor BEAM process message queues and flag processes with mailboxes growing > 500 msgs/sec"

- 1. Samples message queue lengths of all 15,000 active actor processes.
- 2. Discovers bottleneck actor blocked on external synchronous HTTP call.
- 3. Auto-refactors HTTP call to asynchronous cast with correlation ID callback.

Outcome: System-wide message queue cleared from 42,000 to 0 in 1.2s.

Scenario 58: Polyglot Pipeline Orchestrator: Rust Core + Gleam Logic + Python ML

Capabilities: Tagisan Polyglot Harness, Tokio Subprocess Sandboxing

tgs run "Execute hybrid pipeline: Rust ingests sensor data -> Gleam validates rules -> Python computes inference"

- 1. Rust reads 100MB binary sensor stream from shared memory ring buffer.
- 2. Gleam actor applies business validation rules in BEAM runtime.
- 3. Pipes validated records to PyTorch Python script via stdin; returns unified JSON report.

Outcome: Unified polyglot execution completed in 1.4 seconds with zero IPC serialization overhead.

Scenario 59: High-Frequency BEAM Telemetry Metrics Collection and ExDoc Generation

Capabilities: Elixir Telemetry MCP, ExDoc Documentation Engine

tgs run "Attach Telemetry handlers to Phoenix endpoint and generate published HTML API docs with ExDoc"

- 1. Attaches `:telemetry.attach/4` hooks on VM memory, GC runs, and route timings.
- 2. Streams metrics to Prometheus exporter.
- 3. Compiles comprehensive markdown documentation into searchable ExDoc HTML website.

Outcome: Zero-overhead telemetry enabled with published documentation portal.

Scenario 60: Multi-Tenant Actor Partitioning with Isolated Process Heaps

Capabilities: BEAM Actor Memory Isolation, AgentShield

tgs run "Partition 1,000 enterprise tenants into isolated BEAM actor processes with hard RAM quotas"

- 1. Spawns one actor per tenant with dedicated garbage-collected process heap.
- 2. Monitors memory growth using `:erlang.process\_info(pid, :memory)`.
- 3. Safely isolates a runaway tenant generating 2GB RAM without affecting any other tenant processes.

Outcome: True multi-tenant isolation guaranteed by BEAM per-process memory heaps.

Domain 61–75: ■ Data Engineering, Big Data & Real-Time Event Streaming

Scenario 61: Apache Kafka Consumer Group Rebalance Minimization and Partition Tuning

Capabilities: Kafka Admin MCP, AgentShield

tgs run "Analyze rebalance storm on consumer group order-processing and configure cooperative sticky assignor"

- 1. Ingests Kafka broker logs; identifies frequent `CommitFailedException` causing rebalance loops.
- 2. Increases `max.poll.interval.ms` to accommodate heavy batch processing.
- 3. Upgrades partition assignment strategy to `CooperativeStickyAssignor`.

Outcome: Consumer group rebalance downtime eliminated; throughput stabilized at 85,000 msgs/sec.

Scenario 62: Real-Time CDC (Change Data Capture) Ingestion with Debezium to Apache Iceberg

Capabilities: Debezium MCP, Iceberg Catalog Engine

tgs run "Configure Debezium CDC pipeline streaming MySQL binary logs into Apache Iceberg table on MinIO"

- 1. Establishes Debezium MySQL connector tracking table row changes.
- 2. Writes append and update records to Parquet files organized by daily partition.
- 3. Commits snapshot to Apache Iceberg catalog with ACID row-level updates.

Outcome: Sub-5-second data lakehouse freshness achieved with zero source database load.

Scenario 63: Snowflake SQL Query Cost Optimizer & Partition Pruning Accelerator
Capabilities: Snowflake MCP, SQL AST Optimizer
tgs run "Analyze top 10 most expensive Snowflake queries in account and optimize clustering keys"
1. Fetches query profile statistics from `SNOWFLAKE.ACCOUNT_USAGE.QUERY_HISTORY`. 2. Discovers full table scan on 2-billion-row `events` table scanning 1.4 TB per query. 3. Redesigns clustering key to `(event_date, organization_id)` enabling 99.2% partition pruning.
Outcome: Average query runtime reduced from 45s to 1.1s; monthly Snowflake spend cut by 60%.

Scenario 64: DuckDB In-Memory OLAP Vector Processing for Local Gigabyte Datasets
Capabilities: DuckDB Native Engine, Local Ollama
tgs run "Execute analytical aggregations over 50GB Parquet directory using DuckDB vector engine in Rust"
1. Mounts multi-file Parquet directory using DuckDB zero-copy reader. 2. Executes complex multi-stage window aggregations across 8 CPU cores. 3. Emits summarized JSON metrics in 1.8 seconds using under 2GB RAM.
Outcome: Heavy cloud warehouse queries replaced with instant local DuckDB processing.

Scenario 65: Apache Spark Out-Of-Memory (OOM) Skewed Join Remediation
Capabilities: Spark Profiler MCP, Dialectical Code Synthesis
tgs run "Investigate Spark executor OOM error on stage 4 join and apply salting technique to skewed keys"
1. Analyzes Spark UI event timeline; spots 1 executor processing 85% of shuffle data. 2. Identifies key `null` and `default_org` causing severe data skew. 3. Applies key salting with random integer `0..16` to distribute partitions evenly.
Outcome: Spark job completed in 6 minutes with zero executor OOM failures.

Scenario 66: Data Lineage Mapping and GDPR/CCPA Right-to-be-Forgotten Purger
Capabilities: Data Lineage Graph Engine, PostgreSQL MCP
tgs run "Execute verified GDPR erasure request for user_id=9902 across all relational and lakehouse stores"
1. Traverses data lineage graph across PostgreSQL, Redis, Elasticsearch, and S3 Parquet lake. 2. Executes transactional deletes and tombstone markers in transactional stores. 3. Rewrites Parquet files using Iceberg positional delete files to erase historical logs.
Outcome: Cryptographically signed GDPR erasure certificate generated for compliance audit.

Scenario 67: ClickHouse Materialized View Design for Billion-Row Metric Storage
Capabilities: ClickHouse MCP, SQL Optimizer Engine
tgs run "Create SummingMergeTree materialized view in ClickHouse aggregating hourly API telemetry"
1. Creates high-performance `SummingMergeTree` target table partitioned by month. 2. Defines Materialized View aggregating count, errors, and latency quantiles on insert. 3. Verifies dashboard query latency drops from 12 seconds to 8 milliseconds.
Outcome: Billion-row real-time analytics enabled with instant query response.

Scenario 68: Automated Data Quality Gatekeeper: Null Value & Schema Drift Quarantine
Capabilities: Great Expectations Engine, AgentShield
tgs run "Audit incoming customer data CSV against strict schema contract and quarantine corrupt records"
1. Validates 2,000,000 incoming records against Great Expectations JSON suite. 2. Flags 42 records with invalid ISO 8601 timestamps and negative currency amounts. 3. Routes clean records to production Kafka topic; redirects corrupt records to quarantine bucket.
Outcome: Downstream analytics pipeline protected from dirty data corruption.

Scenario 69: Parquet Metadata Inspection and Snappy/Zstd Compression Optimization Capabilities: Parquet Tooling Engine, Rust Arrow Crate tgs run "Benchmark Snappy vs Zstandard (level 7) compression on 100GB access log Parquet dataset" 1. Reads row group metadata and dictionary encodings. 2. Encodes sample dataset using Snappy and Zstandard level 7. 3. Compares metrics: Zstd achieves 38% smaller file size with 12% faster decompression speed on modern CPUs. Outcome: Storage footprint reduced by 38 TB annually across the enterprise.
Scenario 70: dbt (Data Build Tool) Semantic Layer Metric Generation & Test Validation Capabilities: dbt MCP, BigQuery Engine tgs run "Generate dbt semantic layer definitions for Monthly Recurring Revenue (MRR) and run dbt test" 1. Parses SQL models in `models/marts/finance/`. 2. Creates semantic metric definitions for `mrr` and `net_revenue_retention`. 3. Runs `dbt test`; confirms unique and not-null constraints pass across all 12 models. Outcome: Verified semantic metrics deployed to production BI dashboards.
Scenario 71: Redis Cluster Sharding Rebalance and Eviction Policy Hardening Capabilities: Redis Admin MCP, AgentShield tgs run "Rebalance hash slots across 6-node Redis cluster and configure volatile-lru eviction" 1. Checks cluster memory distribution; discovers node 3 at 96% memory capacity. 2. Migrates 2,048 hash slots from node 3 to newly added node 7 with zero connection drops. 3. Sets `maxmemory-policy volatile-lru` preventing unexpected OOM crashes on key bursts. Outcome: Cluster memory utilization balanced evenly at 68% across all nodes.
Scenario 72: Graph Database Modeling in Neo4j for Supply Chain Traversal Capabilities: Neo4j Cypher MCP, Graph Visualization Engine tgs run "Model global semiconductor supply chain in Neo4j and find single points of failure (bottlenecks)" 1. Loads suppliers, manufacturing plants, logistics hubs, and ports as nodes and edges. 2. Executes Cypher centrality queries to compute betweenness centrality scores. 3. Identifies single sub-tier supplier in Taiwan responsible for 92% of critical microcontroller packaging. Outcome: Supply chain vulnerability flagged to executive procurement team with mitigation plan.
Scenario 73: Apache Flink Stateful Stream Windowing for Fraud Velocity Detection Capabilities: Flink Java/Rust Engine, Streaming Event Processor tgs run "Deploy Flink 60-second sliding window detecting > 5 credit card transactions from different cities" 1. Configures Flink keyed stream by `card_number` using event-time watermarking. 2. Computes haversine distance between sequential geolocation transaction coordinates. 3. Triggers immediate fraud lock event when travel speed exceeds 600 mph (impossible travel). Outcome: Card fraud detected and blocked in 42 milliseconds.
Scenario 74: Reverse ETL Pipeline: Syncing BigQuery Data directly into Salesforce CRM Capabilities: BigQuery MCP, Salesforce REST MCP tgs run "Sync high-propensity churn risk scores from BigQuery ML model into Salesforce Account records" 1. Queries BigQuery ML inference view for accounts with churn score > 0.75. 2. Batches 10,000 updates using Salesforce Composite Graph API. 3. Verifies zero rate-limit throttling and updates customer success task queue. Outcome: Account executives alerted to at-risk accounts automatically every morning.

Scenario 75: Automated Data Cataloging and Semantic Tagging via Vector Embeddings

Capabilities: PILOT Vector Memory, Metadata Extraction Engine

tgs run "Crawl 450 database tables and auto-generate business semantic descriptions and PII tags"

1. Scans column names, data types, and sample value distributions.
2. Generates semantic embeddings for each table schema and indexes into vector memory.
3. Tags sensitive PII columns (emails, credit cards, SSNs, phone numbers) with GDPR tags.

Outcome: Data catalog 100% indexed with full-text and semantic search enabled.

Domain 76–90: ■ AI/ML Engineering, Local LLM Inference & Fine-Tuning

Scenario 76: Dual-Brain Inference Routing: Local GGUF for Speed, Cloud for Nuance

Capabilities: Tagisan Dual-Brain Router, Ollama + Gemini 2.5 Pro

tgs run "Analyze user request: if simple formatting use Ollama, if legal contract audit use Gemini Pro"

1. Evaluates complexity score of prompt using local lightweight classifier.
2. Routes basic formatting tasks to local Ollama (0ms latency, zero cloud API cost).
3. Automatically fails over complex 80-page legal indemnification review to Gemini 2.5 Pro.

Outcome: Optimal balance: 82% of queries handled locally for free; complex tasks get frontier reasoning.

Scenario 77: Google Web OAuth Free Frontier Model Routing (gemini-2.5-flash)

Capabilities: Google OAuth Manager, CCPA Internal Endpoint

tgs ask "Explain the mathematical proof of Euler's identity in 3 sentences"

1. Verifies local Google OAuth credentials in `~/config/tagisan/gemini\_oauth.json`.
2. Proactively validates token expiry; auto-refreshes token via Google OAuth refresh grant.
3. Dispatches payload to CCPA endpoint with `antigravity/2.0.0` user agent; streams response.

Outcome: Response received in 1.38s with zero API billing costs.

Scenario 78: Deep Reasoning Problem Solving with gemini-3.1-pro-low

Capabilities: Google OAuth Endpoint, Gemini 3.1 Pro Low

tgs ask -m pro "Synthesize a lock-free multi-producer multi-consumer ring buffer in Rust"

1. Resolves `m pro` alias to `gemini-3.1-pro-low` on Google CCPA gateway.
2. Model activates multi-step internal reasoning/thinking chain.
3. Emits production Rust code with atomic CAS loops and safety invariants.

Outcome: High-complexity algorithms solved with formal verification reasoning in 3.7s.

Scenario 79: Ultra-Low-Latency Assistant Interaction with gemini-2.5-flash-lite

Capabilities: Google OAuth Endpoint, Gemini 2.5 Flash Lite

tgs ask -m lite "Give me 5 synonym verbs for 'accelerate'"

1. Resolves `m lite` alias to `gemini-2.5-flash-lite`.
2. Sends minimal payload directly to edge endpoint.
3. Streams response tokens with first-token latency under 280ms.

Outcome: Instantaneous completion received in 1.02s.

Scenario 80: Hegelian Dialectical Debate for Automated AI Hallucination Elimination

Capabilities: Swarm MoA Debate Engine, 4-Agent Consensus

tgs debate --proposer "Argue that Python is faster than C for matrix math with NumPy" --challenger "Debunk with compiler facts" - Proposer claims Python with NumPy matches C due to BLAS bindings. - Challenger demonstrates boundary overhead, GIL stalls on multi-threading, and non-vectorized custom loops. - Judge reviews cross-examination and rules: Python delegates to C/Fortran, but raw native code wins on cache locality. Outcome: Factually verified consensus synthesized with zero hallucinations.
Scenario 81: Quantizing Raw PyTorch Models into 4-bit GGUF via llama.cpp Capabilities: llama.cpp Toolchain, AgentShield Process Sandbox tgs run "Quantize raw FP16 PyTorch weights in models/qwen/ to Q4_K_M GGUF format" - Converts Safetensors weights to FP16 GGUF intermediate. - Executes `llama-quantize` with `Q4_K_M` block-level quantization matrix. - Validates model perplexity degradation remains under 0.05% while reducing model size from 14GB to 4.2GB. Outcome: Model converted to run on consumer 8GB VRAM GPUs at 68 tokens/sec.
Scenario 82: LoRA (Low-Rank Adaptation) Parameter-Efficient Fine-Tuning for Domain Tasks Capabilities: PyTorch / Unsloth MCP, Python uv Toolchain tgs run "Fine-tune Qwen-2.5-Coder on 5,000 enterprise proprietary API examples using LoRA rank 16" - Tokenizes domain dataset with ChatML template. - Injects trainable LoRA adapter matrices into attention projection layers (\$q, k, v, o\$). - Completes 3 training epochs in 45 minutes; merges adapter into standalone GGUF model. Outcome: Domain model achieves 99.4% accuracy on proprietary internal APIs.
Scenario 83: RAG Pipeline Optimization with Hybrid Sparse/Dense Embedding Retrieval Capabilities: Qdrant Vector MCP, BM25 Tokenizer, Reciprocal Rank Fusion tgs run "Implement hybrid RAG search combining BM25 keyword matching with BGE-m3 dense embeddings" - Computes sparse lexical tokens and dense 1024-dimension vectors in parallel. - Queries Qdrant vector database using reciprocal rank fusion (RRF with \$k=60\$). - Applies Cohere reranker to top 20 candidates; returns top 3 precision passages. Outcome: Retrieval Mean Reciprocal Rank (MRR@10) increased from 0.71 to 0.94.
Scenario 84: Vector Database Index Tuning (HNSW M & efConstruction) in Qdrant Capabilities: Qdrant Admin MCP, Vector Benchmark Engine tgs run "Tune HNSW index parameters on 10M vector collection in Qdrant for < 10ms search latency" - Evaluates recall vs throughput with varying `m` and `ef_construct`. - Reconfigures collection to `m=32`, `ef_construct=256`, and scalar quantization (int8). - Verifies recall stays at 98.6% while memory consumption drops by 75%. Outcome: p99 vector search latency clocked at 7.4 milliseconds.
Scenario 85: Prompt Injection Defense Benchmarking against Red-Team Payloads Capabilities: AgentShield Threat Evaluator, Security Test Suite tgs run "Execute 500 adversarial jailbreak prompts (DAN, Base64, Roleplay, Unicode) against AgentShield" - Dispatches automated battery of indirect and direct prompt injection attacks. - AgentShield AST scanner intercepts attempts to override system instructions. - Intercepts hidden shell execution attempts in returned Markdown links. Outcome: 100% of critical jailbreak and exfiltration payloads intercepted cleanly.
Scenario 86: Semantic Chunking vs. Fixed Window Chunking Document Parser

Capabilities: NLP Parser Engine, Local Embedding Model tgs run "Benchmark semantic similarity boundary chunking against 512-token fixed window on 200 PDFs" 1. Parses document text into sentences. 2. Computes cosine distance between sequential sentence embeddings. 3. Splits chunks when distance exceeds 95th percentile, preserving complete conceptual paragraphs. Outcome: Information fragmentation eliminated; downstream QA accuracy boosted by 28%.
Scenario 87: LLM Token Cost Tracking & Daily Budget Cap Enforcement (\$USD) Capabilities: Tagisan Budget Engine, SQLite Episodic Store tgs run --budget 5.00 "Execute multi-stage code migration across 40 files with hard \$5.00 safety cap" 1. Accurately tracks prompt, completion, and cached tokens across every LLM call. 2. Computes running total using exact provider pricing tables. 3. Automatically halts and alerts user if cumulative spend nears the \$5.00 threshold. Outcome: Zero surprise API bills; financial safety guaranteed by design.
Scenario 88: Serving Ollama Edge Models on Apple Silicon Metal & Linux CUDA Capabilities: Ollama Service Manager, GPU Hardware Profiler tgs run "Inspect GPU layer offloading on Ollama server and optimize num_gpu layers for RTX 4090" 1. Queries Ollama <code>/api/show</code> endpoint to check active VRAM allocation. 2. Detects partial CPU offloading causing 12 tokens/sec bottleneck. 3. Adjusts <code>num_gpu=99</code> and <code>context_length=8192</code> in Modelfile, loading 100% of layers into VRAM. Outcome: Generation speed increased from 12 tokens/sec to 118 tokens/sec.
Scenario 89: Embedding Model Drift Detection & Re-Indexing Workflow Capabilities: PILOT Memory Auditor, Cosine Drift Metric tgs run "Audit vector database for model version drift between text-embedding-ada-002 and text-embedding-3-small" 1. Compares metadata vector dimension signatures across 250,000 collection records. 2. Detects 15,000 records indexed with legacy 1536-dimension embeddings mixed with newer vectors. 3. Triggers automated background re-embedding batch job and rebuilds HNSW index. Outcome: Embedding dimension mismatch resolved with zero query downtime.
Scenario 90: Structured Output Extraction with Strict JSON Schema Guarantees Capabilities: Grammar-Guided LLM Engine, JSON Schema Validator tgs run "Extract financial invoice data into strict JSON matching schemas/invoice.json" 1. Compiles JSON schema into deterministic BNF context-free grammar. 2. Restricts LLM token logits during sampling to only allow syntactically valid JSON tokens. 3. Emits 100% valid JSON payload with zero parsing errors. Outcome: Deterministic structured data extraction achieved on every run.

Domain 91–105: ■ Quantitative Finance, Algorithmic Trading & Risk Control (VELLA)

Scenario 91: High-Frequency Forex Tick Spread Analysis & Slippage Monitoring Capabilities: VELLA Quant Engine, FIX Protocol Parser tgs vella forex --pair "EUR/USD" --bid 1.0842 --ask 1.0844 --lot 10.0 --leverage 50.0

1. Calculates bid-ask spread in pips (0.2 pips) and required margin (\$2,168.40).
2. Computes pip value (\$100.00 per pip) and evaluates liquidity depth across 3 broker feeds.
3. Warns of anomalous spread widening prior to US Non-Farm Payrolls (NFP) announcement.

Outcome: Execution routed to tightest spread ECN liquidity provider, saving \$450 in slippage.

Scenario 92: Value-at-Risk (VaR) Monte Carlo Portfolio Simulation (99% Confidence)

Capabilities: VELLA Monte Carlo Simulator, Rayon Multi-Threading

tgs run "Execute 100,000 Monte Carlo paths for \$5M portfolio over 10-day horizon and calculate 99% VaR"

1. Ingests covariance matrix for 20 asset classes.
2. Generates 100,000 Correlated Gaussian shock paths across multi-core Rayon threads.
3. Computes 99% 10-day Value-at-Risk (\$318,400) and Conditional VaR (Expected Shortfall).

Outcome: Risk report signed and submitted to Chief Risk Officer before market open.

Scenario 93: Real-Time Margin Utilization & Automated Pre-Liquidation De-leveraging

Capabilities: VELLA Risk Controller, Exchange REST API

tgs run "Monitor account margin level; if margin level drops below 120%, close lowest conviction position"

1. Polls equity and margin balance every 500 milliseconds.
2. Detects sudden flash drop in JPY positions dropping margin level to 118%.
3. Issues immediate limit order closing 2 lots of USD/JPY, restoring margin level to 164%.

Outcome: Catastrophic account stop-out liquidation prevented automatically.

Scenario 94: Cross-Exchange Crypto Arbitrage Route Detection with Gas Estimation

Capabilities: Web3 MCP, DEX Liquidity Math Engine

tgs run "Scan Uniswap v3 and Binance ETH/USDT price divergence; calculate net profit after gas & slip"

1. Detects 0.65% price discrepancy between Binance spot orderbook and Uniswap v3 pool.
2. Computes exact Ethereum mainnet gas fee (32 Gwei) and DEX swap fee (0.05%).
3. Confirms net profit of \$1,840; submits flashbot private transaction bundle to avoid front-running.

Outcome: Arbitrage executed profitably on-chain without MEV sandwiching.

Scenario 95: Order Book Imbalance (OBI) Forecasting with Microsecond Telemetry

Capabilities: L2/L3 Orderbook Engine, Rust AVX-512 Vectorization

tgs run "Calculate Order Book Imbalance (OBI) on top 10 levels of BTC-USDT orderbook every 100ms"

1. Ingests live WebSocket L2 orderbook updates.
2. Computes weighted depth imbalance: $OBI = \frac{V_{bid} - V_{ask}}{V_{bid} + V_{ask}}$.
3. Detects institutional spoof wall on bid side pulling liquidity; issues downward price impulse alert.

Outcome: High-frequency trade signals generated with sub-millisecond calculation latency.

Scenario 96: Automated Algorithmic Trailing Stop-Loss Adjustment during Macro Events

Capabilities: VELLA Trade Supervisor, Economic Calendar MCP

tgs run "Tighten trailing stops on all GBP positions to 15 pips 5 minutes before Bank of England rate decision"

1. Tracks global economic calendar countdown.
2. At T-5 minutes, scans active orders and amends broker stop-loss orders via FIX protocol.
3. Locks in \$12,400 in accrued unrealized profit prior to severe rate volatility spike.

Outcome: Capital protected during 120-pip whip-saw macro event.

Scenario 97: FIX Protocol (Financial Information eXchange) Session Parsing & Reconnect

Capabilities: FIX 4.4 Engine, Tokio Network Reconnector

tgs run "Maintain FIX 4.4 session with institutional liquidity provider and handle sequence reset"

1. Manages continuous 30-second Heartbeat messages (`35=0`).
2. Intercepts disconnect; executes Logon (`35=A`) with sequence number resync (`35=4`).
3. Resends missing fill reports without duplicate trade executions.

Outcome: Institutional trading link restored with zero lost trade messages.

Scenario 98: Black-Scholes Greeks Sensitivity Engine (Delta, Gamma, Vega, Theta)

Capabilities: VELLA Options Math Engine, Rust Precision Math

tgs run "Calculate full option Greeks for SPX \$5,000 Call expiring in 14 days with IV=16.5%"

1. Computes \$d_1\$ and \$d_2\$ using Black-Scholes continuous dividend formulation.
2. Calculates Delta (0.54), Gamma (0.0028), Vega (\$14.20), and Theta (-\$3.85/day).
3. Recommends delta-neutral hedge buying 54 shares of underlying index per contract.

Outcome: Accurate option risk parameters delivered instantly.

Scenario 99: Backtesting Mean-Reverting Strategies across 10 Years of M1 Candles

Capabilities: Historical Backtest Engine, DuckDB / Parquet Reader

tgs run "Backtest Bollinger Band mean-reversion strategy on 5,000,000 1-minute GBP/USD candles"

1. Loads 10 years of M1 OHLCV candles from local Parquet storage into memory.
2. Executes vectorized trade simulation accounting for 1.2 pip spread and swap financing.
3. Outputs Sharpe ratio (1.82), Maximum Drawdown (7.4%), and Profit Factor (1.64).

Outcome: 10-year backtest executed in 3.4 seconds with comprehensive equity curve plot.

Scenario 100: Smart Contract Reentrancy Vulnerability Auditing with Slither/Echidna

Capabilities: Solidity AST Parser, Slither MCP, AgentShield

tgs run "Audit contracts/Vault.sol for reentrancy bugs and state update ordering flaws"

1. Parses Solidity abstract syntax tree.
2. Discovers external ether transfer (`msg.sender.call{value: amount}("")`) occurring before state balance reset.
3. Rewrites method to follow Checks-Effects-Interactions pattern and applies OpenZeppelin `ReentrancyGuard`.

Outcome: Critical reentrancy exploit patched before mainnet deployment.

Scenario 101: MEV (Maximal Extractable Value) Sandwich Attack Defense for DEX Swaps

Capabilities: Web3 Mempool Watcher, Slippage Controller

tgs run "Route \$250,000 DAI to USDC swap on Curve using Flashbots RPC with 0.05% slippage cap"

1. Checks public Ethereum mempool for predator sandwich bots.
2. Routes transaction via private Flashbots builder endpoint bypassing public mempool.
3. Sets strict 0.05% slippage tolerance guarantee.

Outcome: Swap executed with \$0 lost to MEV bot extractors.

Scenario 102: Multi-Currency Basket Hedging Strategy Formulation

Capabilities: Correlation Matrix Engine, Swarm MoA Portfolio Team

tgs debate --proposer "Hedge EUR long exposure using USD, CHF, and GBP basket" --challenger "Optimize for lowest carry cost"

1. Evaluates 180-day rolling correlation between EUR/USD, EUR/CHF, and EUR/GBP.
2. Factors in central bank interest rate differentials (carry cost).
3. Formulates optimal basket weightings minimizing tracking error and financing fees.

Outcome: Currency risk hedged with 40% lower carry cost than single-pair hedging.

Scenario 103: Flash Crash Circuit Breaker: Automatic Capital Freezing & Notification

Capabilities: VELLA Circuit Breaker, Telegram / PagerDuty MCP

tgs run "Monitor equity tick velocity; if account loses > 2% in under 60 seconds, cancel all orders and lock" 1. Real-time tick monitor detects sudden 2.4% equity drop during flash crash. 2. Dispatches mass cancel command to all active exchange limit orders. 3. Closes all high-leverage positions and dispatches emergency alert to trading desk via Telegram. Outcome: Account preserved from catastrophic market drawdown.
Scenario 104: Automated Financial News Sentiment Ingestion & Correlation Mapping Capabilities: Bloomberg/Reuters RSS MCP, Gemini 2.5 Flash tgs run "Ingest live financial news stream and compute instant sentiment score for S&P; 500 tech tickers" 1. Ingests breaking news articles via RSS and financial API webhooks. 2. Extracts ticker mentions and evaluates sentiment on a -1.0 to +1.0 polarity scale. 3. Correlates sentiment shifts against real-time orderflow volume spikes. Outcome: Trading desk alerted to breaking sentiment shift 45 seconds ahead of mainstream news.
Scenario 105: Regulatory Trade Reporting Compliance (CFTC / MiFID II) Audit Trail Capabilities: Compliance Ledger Engine, SHA-256 Merkle Tree tgs run "Audit all 42,000 trades executed on 2026-09-14 and compile regulatory MiFID II transaction report" 1. Validates all required regulatory fields: Trader ID, Algorithm ID, UTC Timestamp to microsecond, Price, Volume. 2. Generates SHA-256 Merkle root hash anchoring trade sequence to immutable audit ledger. 3. Exports compliant XML format for submission to Approved Reporting Mechanism (ARM). Outcome: 100% compliant regulatory report compiled and verified without manual auditing.

Domain 106–120: ■ Industrial IoT, SCADA & Cyber-Physical Digital Twins (VELLA)

Scenario 106: Modbus TCP Register Polling & Pressure Relief Valve Telemetry Sync Capabilities: VELLA SCADA Engine, Modbus TCP Protocol tgs vella scada --endpoint "tcp://192.168.1.100:502" --analog 85.4 --disk 74.2 --alarm "trip_cooling" 1. Connects to industrial Modbus PLC; polls holding registers for vessel pressure (85.4 PSI). 2. Compares against safety envelope threshold (80.0 PSI). 3. Automatically triggers emergency cooling auxiliary pump and logs safety trip event. Outcome: Pressure normalized back to 72.0 PSI; chemical explosion risk prevented.
Scenario 107: OPC-UA Industrial Sensor Anomaly Detection in Chemical Refineries Capabilities: OPC-UA Client MCP, Anomaly Detection Model tgs run "Subscribe to 500 OPC-UA sensor nodes in distillation column #2 and detect correlation breakdown" 1. Subscribes to live sensor telemetry streams (temperature, pressure, flow rate). 2. Multivariate anomaly model flags temperature rising while cooling valve reports 100% open. 3. Diagnoses physical valve mechanical seizure; dispatches maintenance work order. Outcome: Faulty valve identified before catalyst bed degradation occurred.
Scenario 108: Digital Twin Thermal Equilibrium Modeling for CNC Machining Centers Capabilities: VELLA Digital Twin Physics Engine, C++ Math Solver tgs run "Simulate spindle thermal expansion on 5-axis CNC mill operating at 18,000 RPM for 4 hours"

- 1. Solves thermal diffusion differential equations across spindle bearing assembly.
- 2. Predicts \$18.4\mu m\$ axial thermal expansion along Z-axis.
- 3. Transmits dynamic G-code tool-length offset compensation to Fanuc CNC controller.

Outcome: Machining tolerance held within \$1\mu m\$ across 4-hour production run.

Scenario 109: Predictive Maintenance: Bearing Vibration FFT Spectral Analysis

Capabilities: Fast Fourier Transform (FFT) Engine, Edge Telemetry

tgs run "Compute 4,096-point FFT on accelerometer timeseries from turbine generator bearing"

- 1. Converts 10 kHz vibration timeseries from time domain to frequency domain.
- 2. Identifies sharp spectral peak at 148 Hz matching Ball Pass Frequency Outer Race (BPFO).
- 3. Estimates remaining useful life (RUL) at 120 operating hours before bearing spalling.

Outcome: Replacement scheduled during routine weekend downtime, avoiding catastrophic turbine shutdown.

Scenario 110: Real-Time PLC (Programmable Logic Controller) State Mirroring

Capabilities: EtherNet/IP & CIP Protocol Engine, VELLA Twin

tgs run "Mirror live Allen-Bradley ControlLogix PLC memory tags into local SQLite digital twin"

- 1. Establishes EtherNet/IP CIP session polling 1,200 PLC tags every 50ms.
- 2. Stores state transitions in local high-speed circular memory buffer.
- 3. Detects asynchronous interlock race condition between conveyor belt and robotic arm.

Outcome: Interlock bug diagnosed and patched in ladder logic in 15 minutes.

Scenario 111: Electric Grid Load Balancing and Transformer Overheat Prevention

Capabilities: Smart Grid Protocol Engine, Swarm MoA

tgs run "Analyze 12 substation transformer loads during heatwave and re-route feeder lines"

- 1. Ingests oil temperature and apparent power (kVA) telemetry across 12 distribution substations.
- 2. Discovers Substation B transformer operating at 108% rated capacity with oil temp at 98°C.
- 3. Issues SCADA tie-switch closing commands transferring 4.2 MW load to adjacent Substation C.

Outcome: Transformer temperature stabilized at 82°C, avoiding residential blackout.

Scenario 112: HVAC Energy Efficiency Optimization in Multi-Story Smart Buildings

Capabilities: BACnet MCP, Thermodynamic Energy Model

tgs run "Optimize chiller plant staging and VAV dampers across 40-story office building based on weather forecast"

- 1. Connects to building automation system via BACnet/IP protocol.
- 2. Pulls solar irradiance forecast and occupancy sensor counts.
- 3. Pre-cools building during off-peak electricity hours (\$0.06/kWh); reduces chiller load during peak hours (\$0.28/kWh).

Outcome: Building monthly energy cost reduced by 22.4% without compromising tenant comfort.

Scenario 113: Industrial Water Treatment Facility Turbidity & pH Feedback Loops

Capabilities: PID Controller Engine, Water Quality Sensors

tgs run "Monitor incoming stormwater runoff turbidity and adjust coagulant chemical dosing pumps"

- 1. Detects sudden turbidity surge from 12 NTU to 180 NTU following heavy rainfall.
- 2. Automatically scales polyaluminum chloride (PAC) dosing pump speed via 4-20mA analog output.
- 3. Modulates caustic soda injection to maintain effluent pH strictly between 7.2 and 7.6.

Outcome: Treated water purity maintained 100% within EPA regulatory drinking standards.

Scenario 114: Factory Floor AGV (Automated Guided Vehicle) Collision Avoidance Mesh

Capabilities: ROS2 (Robot Operating System) Bridge, Dijkstra Mesh

tgs run "Calculate collision-free routing paths for 18 autonomous warehouse forklifts"

1. Ingests real-time LIDAR SLAM coordinates of all 18 automated guided vehicles.
2. Detects path conflict at aisle intersection 4 between AGV-03 and AGV-09.
3. Dynamically assigns priority yield token to AGV-03 and computes alternate detour for AGV-09.

Outcome: Zero factory collisions; continuous warehouse pick-and-pack throughput maintained.

Scenario 115: Smart Meter Telemetry Aggregation over Cellular LTE-M / NB-IoT

Capabilities: MQTT-SN / CoAP Protocol Engine, TimeSeries Store

tgs run "Ingest hourly electricity consumption packets from 250,000 smart meters over MQTT broker"

1. Connects to distributed EMQX MQTT cluster subscribing to `meters+/consumption`.
2. Decompresses CBOR-encoded binary payloads and validates digital signature.
3. Writes 250,000 metrics to ClickHouse in micro-batches every 2 seconds.

Outcome: Million-meter ingestion pipeline operates on under 4 CPU cores.

Scenario 116: Pipeline Leak Detection using Acoustic Sensor Correlation Arrays

Capabilities: Acoustic Signal Processing, Cross-Correlation Solver

tgs run "Correlate acoustic hydrophone data along 50km oil pipeline to pinpoint rupture location"

1. Ingests high-frequency acoustic wave sensors located at 5km intervals.
2. Computes time-difference-of-arrival (TDOA) cross-correlation between sensor 4 and sensor 5.
3. Pinpoints pinhole leak at kilometer marker 23.415 with accuracy within ± 10 meters.

Outcome: Pipeline emergency shutoff valves closed; environmental spill minimized to under 5 gallons.

Scenario 117: Wind Turbine Pitch Control Optimization in High-Wind Gusts

Capabilities: Aerodynamic Model, High-Speed PLC Interface

tgs run "Modulate blade pitch angle on 3.5 MW wind turbine to prevent rotor overspeed in 65 mph gusts"

1. Anemometer telemetry reports sudden 65 mph wind gust approaching turbine rotor.
2. Computes aerodynamic lift-drag polar equations.
3. Feathers blade pitch angle from 4° to 18° within 1.2 seconds, limiting generator RPM to safety rating.

Outcome: Turbine kept online generating clean power without mechanical brake stress.

Scenario 118: Solar Inverter Efficiency Tracking and MPPT Fault Isolation

Capabilities: Solar MPPT Engine, Modbus SunSpec Protocol

tgs run "Audit 40 solar string inverters across 50 MW farm and detect degraded photovoltaic strings"

1. Polls SunSpec Modbus registers for DC voltage, current, and AC power output.
2. Normalizes output against ambient temperature and horizontal pyranometer irradiance.
3. Identifies String 14B underperforming by 42%; diagnoses failed bypass diode.

Outcome: Defective string repaired, restoring \$18,000 in monthly lost solar energy.

Scenario 119: Emergency Industrial SCADA Air-Gap Isolation Protocol

Capabilities: AgentShield Cyber Defense, Industrial Firewall MCP

tgs run "Detect unauthorized external IP connection on SCADA subnet and execute immediate network air-gap"

1. Network monitoring agent detects rogue SSH outbound connection from HMI machine to Russian IP.
2. AgentShield immediately trips Moxa industrial managed switch port into shutdown.
3. Isolates OT network from IT network completely while keeping local safety PLC loops operational.

Outcome: SCADA network successfully air-gapped; zero plant equipment compromise.

Scenario 120: Cold-Chain Pharmaceutical Temperature Logger Excursion Triaging

Capabilities: IoT BLE Telemetry Engine, FDA 21 CFR Part 11 Audit

tgs run "Audit temperature logs from shipment of mRNA vaccines and verify cold-chain compliance (-80°C)"

- 1. Downloads cryogenic temperature logger data across 72-hour international flight transit.
- 2. Detects single 14-minute temperature rise from -82°C to -74°C during dry-ice replenishment.
- 3. Compares against manufacturer stability data; validates that thermal excursion remained within allowable bounds.

Outcome: Vaccine batch certified safe for clinical administration with complete FDA audit certificate.

Domain 121–135: ■■ Aerospace, Satellite Telemetry & Defense Systems (VELLA)

Scenario 121: Low Earth Orbit (LEO) Satellite SGP4 TLE Orbit Propagation

Capabilities: VELLA Aerospace Engine, SGP4 Orbit Solver

```
tgs vella aerospace --minutes 90.0 --tle "1 25544U 98067A 26258.51460395 .00016717 00000-0 10270-3 0 9018\n2 25544 51.6416 247.4627 0006703 130.5360 325.0288 15.72125391563537"
```

- 1. Parses NORAD Two-Line Element (TLE) format for the International Space Station.
- 2. Executes SGP4 perturbation model accounting for Earth oblateness (\$J_2, J_3, J_4\$) and atmospheric drag.
- 3. Computes exact ECI state vectors (\$X, Y, Z\$) and ground track latitude/longitude after 90 minutes.

Outcome: Orbit propagated with sub-meter numerical precision.

Scenario 122: Ground Station Pass Visibility Window and Antenna Azimuth/Elevation

Capabilities: Orbital Geometry Engine, Ground Station MCP

```
tgs run "Calculate next 24-hour pass windows and antenna Az/El tracking angles for Svalbard ground station"
```

- 1. Evaluates satellite position relative to Svalbard ground station coordinates (\$78.22^\circ\text{N}, 15.40^\circ\text{E}\$).
- 2. Filters passes with minimum elevation angle $> 10^\circ$ above horizon.
- 3. Generates 6 daily pass schedules with Acquisition of Signal (AOS), Maximum Elevation, and Loss of Signal (LOS).

Outcome: Ground station antenna tracking angles exported to auto-tracker controller.

Scenario 123: Satellite Battery Depth-of-Discharge (DoD) Thermal Modeling

Capabilities: Spacecraft Power Simulator, VELLA Aerospace

```
tgs run "Model Lithium-Ion battery state-of-charge through 14 orbital eclipse cycles of 36 minutes each"
```

- 1. Calculates solar panel power generation in sunlight and zero generation during eclipse.
- 2. Computes power drain from payload instruments, ADCS reaction wheels, and avionics (140W).
- 3. Verifies battery Depth-of-Discharge remains below 28%, preserving 10-year battery mission life.

Outcome: Power budget validated; heater duty cycle optimized to prevent battery freezing.

Scenario 124: Orbital Conjunction Assessment & Collision Avoidance Maneuver Planning

Capabilities: Conjunction Assessment Engine, Swarm MoA

```
tgs run "Analyze Space-Track CDM (Conjunction Data Message); miss distance is 142m against space debris"
```

- 1. Ingests CDM covariance ellipsoids; calculates probability of collision ($P_c = 4.8 \times 10^{-3}$, above 10^{-4} threshold).
- 2. Formulates impulsive ΔV burn maneuver vector: 0.18 m/s along velocity vector.
- 3. Re-propagates orbits confirming miss distance increases to 4.8 km with zero secondary conjunctions.

Outcome: Thruster burn sequence approved and transmitted to satellite on next uplink pass.

Scenario 125: Spacecraft Attitude Determination & Control System (ADCS) Gyro Drift

Capabilities: Extended Kalman Filter (EKF), Quaternion Math

```
tgs run "Filter noisy star tracker and MEMS gyroscope telemetry to estimate spacecraft attitude quaternion"
```

- 1. Ingests 100 Hz star tracker quaternions and angular rate measurements.
- 2. Implements 7-state Multiplicative Extended Kalman Filter (MEKF).
- 3. Estimates and subtracts gyroscope bias drift, locking spacecraft pointing accuracy to 0.02°.

Outcome: Satellite optical payload stays precisely locked onto terrestrial target.

Scenario 126: Satellite Solar Array Sun-Tracking Angle Optimization

Capabilities: Orbital Kinematics Engine, VELLA Aerospace

tgs run "Calculate solar array drive mechanism (SADM) rotation angle maximizing solar incidence angle"

- 1. Computes Sun vector in satellite body coordinate frame throughout orbit.
- 2. Formulates single-axis SADM tracking angle minimizing cosine loss.
- 3. Increases power generation by 31% compared to fixed-angle orientation.

Outcome: Power generated sufficient to operate payload in continuous observation mode.

Scenario 127: Telemetry Decommulation: CCSDS Packet Framing and Checksum Validation

Capabilities: CCSDS Space Packet Parser, Bit Manipulation Engine

tgs run "Parse 50MB raw binary downlink stream into CCSDS space packets and extract instrument telemetry"

- 1. Synchronizes onto 32-bit sync word `0x1ACFFC1D` (ASM).
- 2. Validates Reed-Solomon $(255, 223)$ forward error correction and CRC-16 checksums.
- 3. Decommutes 4,200 telemetry channels (voltages, temperatures, payload data) into structured SQLite.

Outcome: 100% telemetry recovered with zero corrupted frame drops.

Scenario 128: Atmospheric Re-entry Trajectory Simulation and Heat Shield Stress

Capabilities: Aerodynamic Entry Solver, High-Order Runge-Kutta

tgs run "Simulate 4th-order Runge-Kutta atmospheric re-entry from 120km to splashdown at Mach 25"

- 1. Integrates 3-DOF equations of motion through 1976 Standard Atmosphere.
- 2. Computes stagnation point convective heat flux using Sutton-Graves formulation.
- 3. Confirms maximum deceleration remains under 7.8 Gs and thermal protection tile stress is within limits.

Outcome: Re-entry trajectory verified safe for capsule recovery.

Scenario 129: Deep Space Optical Communications Link Budget Calculation

Capabilities: Link Budget Engine, Laser Physics

tgs run "Calculate optical laser communication link budget from Mars orbit (1.5 AU) to Earth ground telescope"

- 1. Calculates free-space path loss at 1550nm wavelength over 2.25×10^8 km ($L_p = 295$ dB).
- 2. Factors in 5W laser transmitter, 22cm spacecraft telescope, and 5m Earth receiver telescope.
- 3. Demonstrates positive link margin (+4.2 dB) supporting 25 Mbps data downlink.

Outcome: High-definition video transmission from Mars orbit proven feasible.

Scenario 130: Drone Swarm Decentralized Mesh Relay and Jamming Detection

Capabilities: Mesh Routing Engine, RF Spectrum Analyzer MCP

tgs run "Coordinate ad-hoc 802.11s mesh network across 12 autonomous UAVs under GPS jamming"

- 1. Detects GPS spoofing/jamming on 3 forward reconnaissance drones.
- 2. Switches navigation to visual-inertial odometry (VIO) and relative range-bearing mesh.
- 3. Reroutes video telemetry through adjacent non-jammed drone relays to ground command.

Outcome: Drone swarm mission continued successfully with zero dropped video feeds.

Scenario 131: Avionics ARINC 429 Bus Message Decoding and Parity Checking

Capabilities: ARINC 429 Protocol Engine, Binary Parser

tgs run "Decode 32-bit ARINC 429 words from flight control computer and verify odd parity"

1. Extracts Label (bits 1-8), Source/Destination Identifier, Data Field, Sign/Status Matrix, and Parity bit. 2. Validates odd parity on word 203 (Selected Altitude: 34,000 ft). 3. Rejects 2 corrupted words caused by electromagnetic lightning discharge interference. Outcome: Flight computer data bus filtered cleanly with zero false autopilot commands.
Scenario 132: Radiation SEU (Single-Event Upset) Memory Bit-Flip Error Scrubbing Capabilities: EDAC (Error Detection and Correction) Simulator, Spacecraft OS tgs run "Simulate cosmic ray bit-flip in flight software RAM and verify Triple Modular Redundancy (TMR)" 1. Injects hardware bit-flip into critical thruster firing duration register. 2. Triple Modular Redundancy (TMR) voting logic compares 3 independent memory copies. 3. Majority voting circuit (2 out of 3) catches discrepancy, corrects bit, and logs radiation event. Outcome: Flight software execution continued with zero thruster misfire.
Scenario 133: CubeSat Power Budget Allocation under Eclipse Conditions Capabilities: CubeSat Systems Engineering Model, VELLA Aerospace tgs run "Balance 3U CubeSat power states: payload, UHF beacon, attitude reaction wheels" 1. Analyzes energy state across 90-minute orbit. 2. Determines that keeping hyperspectral camera on during eclipse depletes battery past 50% limit. 3. Adjusts state machine schedule: powers down camera 2 minutes prior to orbital sunset. Outcome: CubeSat power margin stabilized at +18%.
Scenario 134: Missile Warning Radar Doppler Shift Trajectory Estimation Capabilities: Radar Signal Processing, Kalman Tracking Filter tgs run "Track hypersonic glide vehicle trajectory from radar return Doppler pulses and estimate impact point" 1. Processes pulsed Doppler radar returns measuring range, azimuth, and Doppler velocity. 2. Applies Unscented Kalman Filter (UKF) to non-ballistic atmospheric skipping trajectory. 3. Computes estimated impact ellipse 8 minutes prior to terminal descent. Outcome: Early warning interceptor trajectory calculated and queued.
Scenario 135: Geosynchronous Satellite Station-Keeping Fuel Depletion Forecast Capabilities: Orbital Maneuver Math, Hydrazine Fuel Engine tgs run "Calculate remaining delta-V and mission lifetime for GEO satellite using 12.4 kg remaining hydrazine" 1. Computes annual station-keeping ΔV requirements: North-South (48 m/s/yr), East-West (2 m/s/yr). 2. Applies Tsiolkovsky rocket equation with monopropellant thruster $I_{sp} = 220 \text{ s}$. 3. Forecasts remaining operational lifetime: 3.4 years, reserving 2.1 kg for final graveyard orbit disposal. Outcome: End-of-life deorbit plan scheduled compliant with IADC space debris guidelines.

Domain 136–150: ■ Bioinformatics, Healthcare & Genomic Analysis (VELLA)

Scenario 136: Next-Generation Sequencing (NGS) FASTQ Quality Filtering & Trimming Capabilities: VELLA Bio Engine, High-Speed String Matcher tgs run "Process 10,000,000 paired-end FASTQ reads; trim Illumina adapters and filter reads with Phred Q < 30" 1. Ingests raw <code>.fastq.gz</code> files using streaming decompression. 2. Trims TruSeq adapter sequences using sliding-window algorithm. 3. Filters out reads with average Phred quality score below Q30 (99.9% base accuracy). Outcome: Cleaned reads ready for downstream variant calling with 99.2% alignment efficiency.

Scenario 137: FASTA Global and Local Sequence Alignment (Needleman-Wunsch / Smith-Waterman)**Capabilities:** VELLA Bio Engine, SIMD Dynamic Programming

```
tgs vella bio --target "ACTGATCGATCGATCG" --template "ACTGATCGTTCGATCG" --ref-genome "GRCh38"
```

1. Implements Smith-Waterman local alignment matrix with affine gap penalties.
2. Computes optimal alignment score (14 matches, 1 mismatch, 0 gaps).
3. Identifies single nucleotide polymorphism (SNP) at position 9: Cytosine substituted by Thymine (C>T).

Outcome: Exact alignment coordinates and substitution identified in 1.4 milliseconds.**Scenario 138: Variant Call Format (VCF) Parsing and Rare Pathogenic Mutation Annotation****Capabilities:** VCF Parser Engine, ClinVar / dbSNP MCP

```
tgs run "Filter patient whole-exome VCF for de novo non-synonymous mutations in cardiomegaly genes"
```

1. Parses 4.2 million variant rows from patient VCF file.
2. Filters for protein-altering missense and nonsense variants with allele frequency < 0.001 in gnomAD.
3. Cross-references ClinVar database; flags pathogenic mutation in `MYH7` gene (p.Arg403Gln).

Outcome: Genetic cause of hypertrophic cardiomyopathy identified for genetic counselor.**Scenario 139: CRISPR-Cas9 On-Target and Off-Target Cleavage Probability Scoring****Capabilities:** CRISPR Guide RNA Engine, Machine Learning Scorer

```
tgs run "Design 20nt sgRNA targeting exon 3 of BCL11A and compute CFD off-target cleavage scores"
```

1. Identifies 20nt guide sequences adjacent to SpCas9 PAM site (5'-NGG-3').
2. Evaluates on-target cutting efficiency using Doench Rule Set 2 score (88.4).
3. Scans reference genome for off-target sites; validates zero off-target sites with Cutting Frequency Determination (CFD) score > 0.02.

Outcome: Optimal sgRNA candidate exported for therapeutic sickle-cell gene editing.**Scenario 140: Single-Cell RNA-Seq Expression Matrix Clustering & Cell Typing****Capabilities:** Single-Cell Transcriptomics Engine, PCA/UMAP Solver

```
tgs run "Cluster 20,000 peripheral blood mononuclear cells (PBMCs) and annotate T-cell and B-cell subsets"
```

1. Normalizes single-cell count matrix; selects top 2,000 highly variable genes.
2. Executes Principal Component Analysis (PCA) and computes UMAP 2D projection.
3. Identifies cell clusters using canonical marker genes: CD3E (T-cells), CD19 (B-cells), CD14 (Monocytes).

Outcome: High-resolution cell atlas generated with automated cell proportion report.**Scenario 141: Protein Secondary Structure Prediction from Amino Acid Sequences****Capabilities:** Protein Biophysics Engine, Transformer Model

```
tgs run "Predict alpha-helix, beta-sheet, and coil propensity for 450-residue kinase enzyme"
```

1. Tokenizes amino acid sequence.
2. Evaluates Chou-Fasman and GOR conformational parameter weights.
3. Annotates catalytic ATP-binding pocket and active site aspartate residue with secondary structure coordinates.

Outcome: 3D structure prediction verified against AlphaFold DB.**Scenario 142: Pharmacogenomics: Drug-Gene Interaction Screening (CYP450 Metabolism)****Capabilities:** CPIC Clinical Guidelines Engine, Patient Genome Parser

```
tgs run "Screen patient CYP2D6 and CYP2C19 star alleles against CPIC guidelines for Clopidogrel dosing"
```

1. Identifies patient genotype: CYP2C19 *2/*2 (poor metabolizer).
2. Evaluates clinical pharmacogenomics guidelines (CPIC).
3. Alerts physician: Patient cannot bioactivate Clopidogrel (Plavix); recommends alternative antiplatelet (Prasugrel/Ticagrelor).

Outcome: Adverse cardiovascular event avoided through individualized genomic medicine.

Scenario 143: Bacterial Antibiotic Resistance Gene Identification (AMR Profiling)**Capabilities:** CARD Database Engine, HMMER Protein Domain Matcher

tgs run "Scan assembled Klebsiella pneumoniae genome for beta-lactamase and carbapenemase resistance genes"

1. Translates genomic open reading frames into protein sequences.
2. Queries Comprehensive Antibiotic Resistance Database (CARD) using profile HMMs.
3. Identifies presence of `blaKPC-2` (KPC carbapenemase), indicating resistance to carbapenems.

Outcome: Infection control hospital team alerted to carbapenem-resistant enterobacteriaceae.**Scenario 144: Cancer Driver Gene Mutation Enrichment Analysis****Capabilities:** Oncogenomics Engine, Fisher Exact Test Solver

tgs run "Perform driver gene enrichment on somatic mutation callset from 50 glioblastoma tumor samples"

1. Separates somatic tumor mutations from matched germline blood samples.
2. Calculates background mutation rate per megabase.
3. Computes statistically significant non-synonymous enrichment in `EGFR`, `PTEN`, and `TP53` ($p < 10^{-8}$).

Outcome: Key oncogenic driver pathways highlighted for targeted kinase inhibitor therapy.**Scenario 145: Phylogenetic Tree Reconstruction from Multiple Sequence Alignments****Capabilities:** Phylogenetic Engine, Maximum Likelihood Solver

tgs run "Construct maximum-likelihood phylogenetic tree for 40 viral spike glycoprotein sequences"

1. Aligns sequences using Clustal Omega algorithm.
2. Evaluates optimal nucleotide substitution model (GTR+G+I).
3. Reconstructs rooted phylogenetic tree with 1,000 bootstrap replicates and exports Newick format.

Outcome: Viral lineage divergence timeline and evolutionary clade branching mapped.**Scenario 146: High-Throughput Ligand-Protein Docking Affinity Scoring****Capabilities:** AutoDock Vina Engine, Chemical Structure Parser

tgs run "Screen 5,000 small molecule ligands against SARS-CoV-2 main protease active binding pocket"

1. Prepares protein PDBQT receptor grid centered on catalytic dyad (Cys145, His41).
2. Executes grid-based conformational docking search with AutoDock Vina scoring function.
3. Ranks top 10 compounds exhibiting binding affinity lower than -9.5 kcal/mol.

Outcome: Lead therapeutic candidates isolated for in-vitro wet lab testing.**Scenario 147: Genome-Wide Association Study (GWAS) Manhattan Plot Outlier Extraction****Capabilities:** Statistical Genetics Engine, PLINK MCPtgs run "Process GWAS association results across 8,000,000 SNPs and extract genome-wide significant loci ($p < 5 \times 10^{-8}$)"

1. Reads logistic regression p-values from case-control study.
2. Calculates genomic inflation factor ($\lambda_{GC} = 1.02$, confirming zero population stratification).
3. Identifies 4 novel lead SNPs on chromosome 6 within the HLA region.

Outcome: Manhattan plot coordinates and risk allele odds ratios compiled into publication tables.**Scenario 148: Electronic Health Record (EHR) De-identification for HIPAA Compliance****Capabilities:** AgentShield Medical Guard, Named Entity Recognition (NER)

tgs run "Scrub 10,000 unstructured clinical nursing notes of all 18 HIPAA Safe Harbor identifiers"

1. Executes medical NER model detecting patient names, dates, hospital names, phone numbers, and MRNs.
2. Replaces identifiers with consistent synthetic pseudonyms (`[PATIENT\_A]`, `[DATE\_OFFSET\_14]`).
3. Verifies zero leakage using secondary adversarial auditing agent.

Outcome: 100% HIPAA-compliant research dataset created for multi-institutional research.

Scenario 149: Clinical Trial Cohort Inclusion/Exclusion Criteria Automated Matching

Capabilities: Clinical NLP Engine, FHIR Patient API MCP

```
tgs run "Match oncology clinic patient roster against ClinicalTrials.gov NCT04285268 eligibility criteria"
```

1. Connects to hospital Fast Healthcare Interoperability Resources (FHIR) API.
2. Evaluates inclusion criteria: Age 18-75, Stage IV NSCLC, EGFR exon 19 deletion, ECOG PS 0-1.
3. Checks exclusion criteria: No previous treatment with 3rd-generation TKI.

Outcome: 12 eligible clinical trial candidate patients matched and routed to primary oncologists.

Scenario 150: Epidemic SIR (Susceptible-Infectious-Recovered) Vector Spread Modeling

Capabilities: Epidemiology ODE Solver, VELLA Bio

```
tgs run "Simulate viral outbreak across city of 1,500,000 with basic reproduction number $R_0 = 2.8$"
```

1. Solves system of non-linear ordinary differential equations: $\frac{dS}{dt}$, $\frac{dI}{dt}$, $\frac{dR}{dt}$.
2. Factors in non-pharmaceutical interventions (NPI) reducing contact rate β by 40% on day 14.
3. Projects peak ICU bed demand and calculates critical vaccination threshold required for herd immunity (64.3%).

Outcome: Municipal pandemic response strategy generated and delivered to public health authorities.

Module 11: Capstone Certification Labs & Practical Exams

To achieve Tagisan Certified Architect (TCA) status, complete these 3 hands-on capstone labs:

- **Lab 1: Autonomous Self-Healing Code Pipeline**

Configure Tagisan to monitor a git repository, detect compilation failures, execute a 3-round Hegelian debate between a Refactoring Agent and a Security Auditor, apply the fixes, and verify that cargo test passes.

- **Lab 2: Enterprise Market Intelligence Swarm**

Deploy a multi-agent swarm orchestrating Brave Search MCP, PostgreSQL MCP, and Qdrant Vector Memory to monitor financial regulatory filings, compute sentiment shifts, and index insights with Reciprocal Rank Fusion.

- **Lab 3: Air-Gapped Cyber Defense Cluster**

Deploy Tagisan with zero external network connectivity (TAGISAN_LOCAL_ONLY=1), local Ollama Qwen2.5-Coder, and local SQLite memory. Verify that AgentShield successfully intercepts all 8 simulated exfiltration attacks in the brutal test suite.

Master Certification Checklist:

- [✓] Build and verify native tgs release binary.
- [✓] Connect Google Web OAuth (tgs auth login gemini) and test -m pro and -m lite.
- [✓] Verify graceful offline fallback to local Ollama on logout.
- [✓] Execute a multi-round Hegelian Dialectical Debate.
- [✓] Install and execute an external MCP tool from the 500 catalog.
- [✓] Implement a custom RFC-004 skill under .ecc/skills/.
- [✓] Successfully execute at least 5 scenarios from Module 10.
- [✓] Pass all 8 brutal security and transport tests in the test suite.